

Problem Set 9
Due: Apr. 26, 2021

1. Section 4.1 Problem 13.

Consider a PDF that is positive only within an interval $[a, b]$ and is symmetric around the mean $(a + b)/2$. Let X and Y be independent random variables that both have this PDF. Suppose that you have calculated the PDF of $X + Y$. How can you easily obtain the PDF of $X - Y$?

2. Section 4.2 Problem 18.

Consider four random variables, W, X, Y, Z , with

$$\mathbf{E}[W] = \mathbf{E}[X] = \mathbf{E}[Y] = \mathbf{E}[Z] = 0$$

$$\text{var}(W) = \text{var}(X) = \text{var}(Y) = \text{var}(Z) = 1$$

and assume that W, X, Y, Z are pairwise uncorrelated. Find the correlation coefficients $\rho(R, S)$ and $\rho(R, T)$, where $R = W + X$, $S = X + Y$, and $T = Y + Z$.

3. Let $\{X_r : r \geq 1\}$ be independent exponential random variables with respective parameters $\{\lambda_r : r \geq 1\}$ no two of which are equal. Find the density function of $S_n = \sum_{r=1}^n X_r$. [Hint: Use induction.]
4. Let X and Y have a bivariate normal density with zero means, variances σ^2, τ^2 , and correlation ρ . Show that:

(a) $\mathbf{E}[X|Y] = \frac{\rho\sigma}{\tau}Y$,

(b) $\text{var}[X|Y] = \sigma^2(1 - \rho^2)$,

(c) $\mathbf{E}[X|X + Y = z] = \frac{(\sigma^2 + \rho\sigma\tau)z}{\sigma^2 + 2\rho\sigma\tau + \tau^2}$,

(d) $\text{var}[X|X + Y = z] = \frac{\sigma^2\tau^2(1 - \rho^2)}{\sigma^2 + 2\rho\sigma\tau + \tau^2}$.